

```
In [1]: # Sample log data (you can also read from file)
logs = [
    "2026-04-10 10:00:01 INFO User login successful",
    "2026-04-10 10:01:15 ERROR Database connection failed",
    "2026-04-10 10:02:20 WARN Disk space low",
    "2026-04-10 10:03:10 INFO File uploaded",
    "2026-04-10 10:04:05 ERROR Timeout occurred"
]

# -----
# MAP PHASE
# -----
mapped = []

for line in logs:
    parts = line.split()
    if len(parts) >= 3:
        level = parts[2]
        mapped.append((level, 1))

print("Mapped Output:")
print(mapped)

# -----
# SHUFFLE + SORT PHASE
# -----
shuffle = {}

for key, value in mapped:
    if key not in shuffle:
        shuffle[key] = []
    shuffle[key].append(value)

print("\nShuffled Output:")
print(shuffle)

# -----
# REDUCE PHASE
# -----
reduced = {}

for key in shuffle:
    reduced[key] = sum(shuffle[key])

print("\nFinal Output (Log Counts):")
for key, value in reduced.items():
    print(key, value)
```

Mapped Output:

```
[('INFO', 1), ('ERROR', 1), ('WARN', 1), ('INFO', 1), ('ERROR', 1)]
```

Shuffled Output:

```
{'INFO': [1, 1], 'ERROR': [1, 1], 'WARN': [1]}
```

Final Output (Log Counts):

```
INFO 2
```

```
ERROR 2
```

```
WARN 1
```

In []: